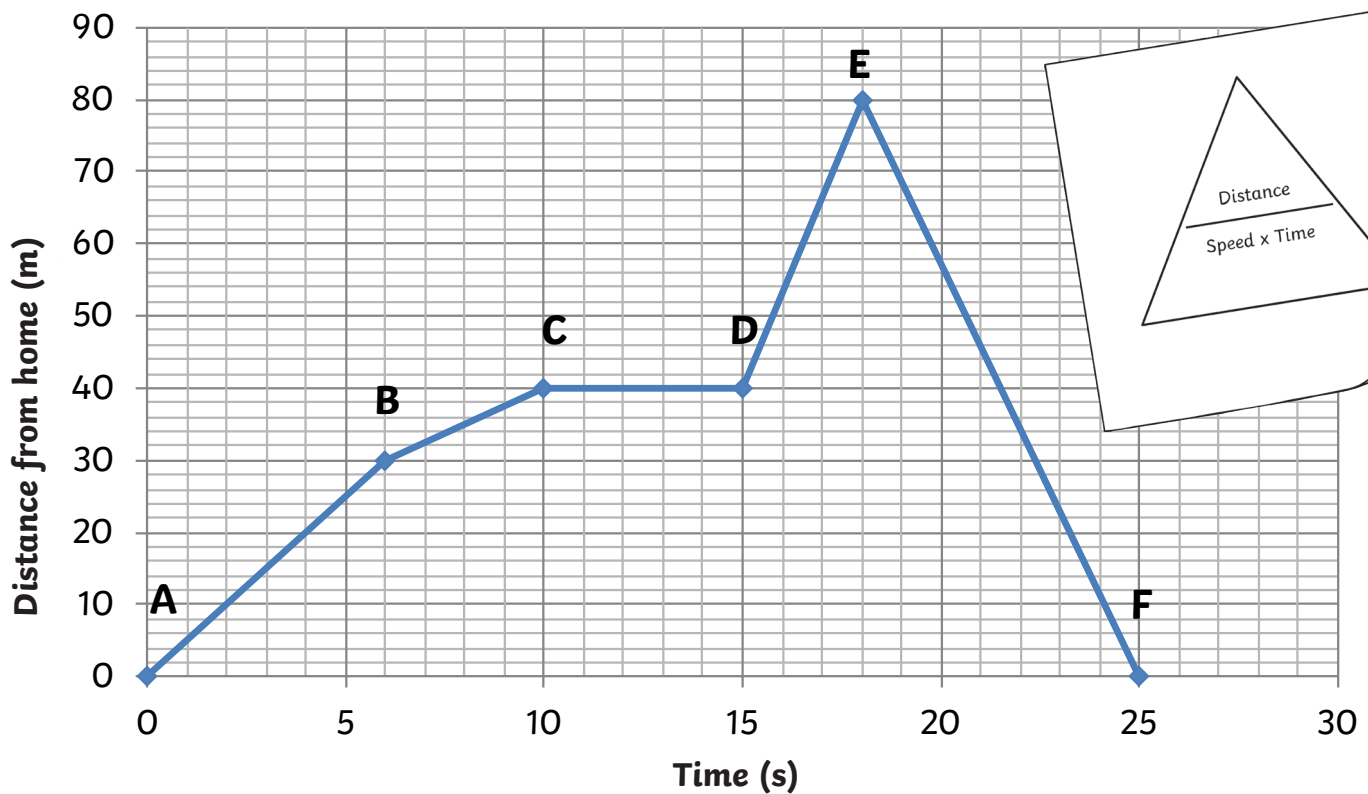


Distance Versus Time Graph



The graph shows the journey of a cyclist. He leaves his starting position heading east.

1. How far has the cyclist traveled between points A and B?

2. How far did the cyclist travel throughout his entire journey?

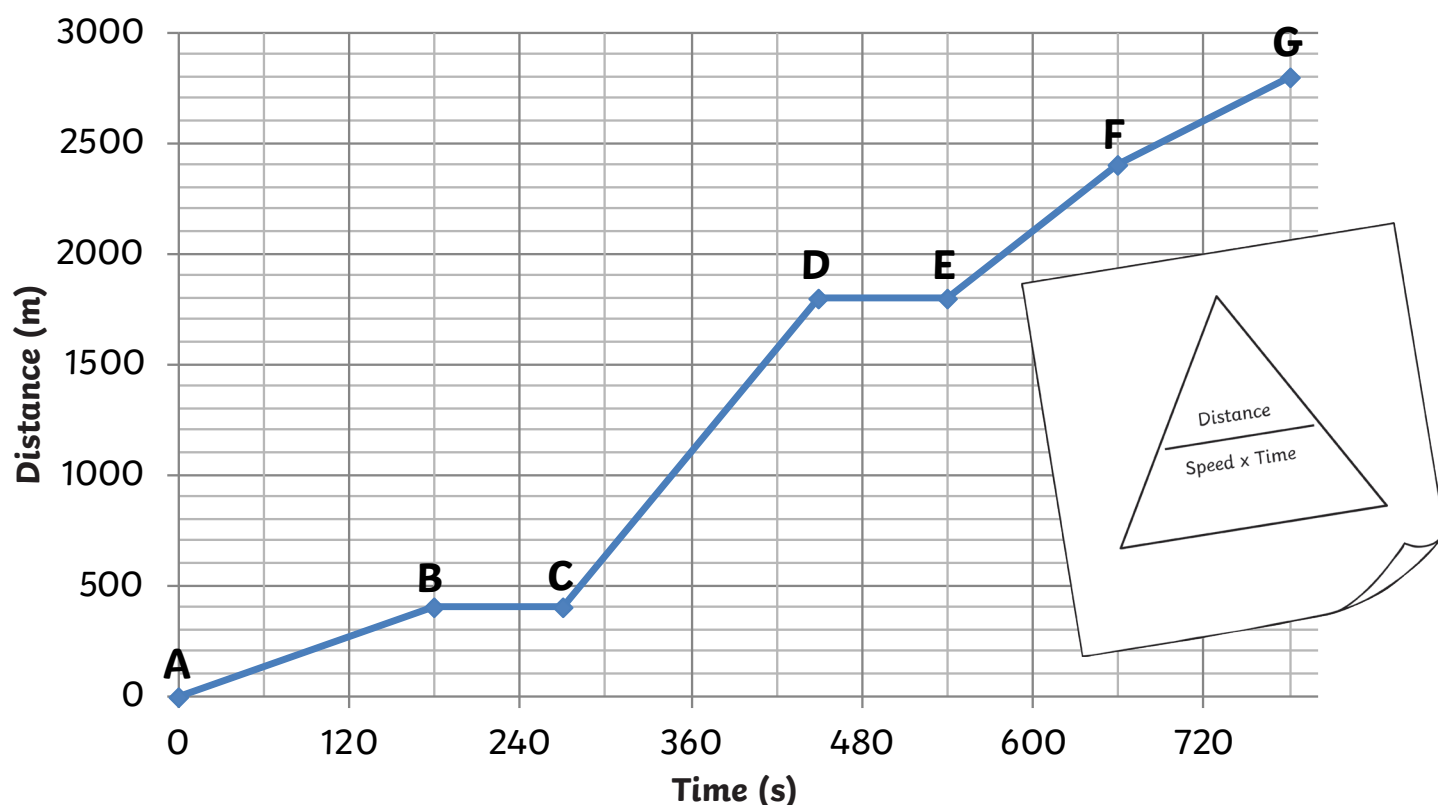
3. Describe the cyclist's speed between points C and D.

4. Describe the cyclist's speed between points D and E.

5. Describe the cyclist's speed between points B and C.

6. Between which two points is the cyclist traveling the fastest?
7. Between which two points is the cyclist traveling the slowest?
8. Calculate the average speed of the cyclist between points A and E.
9. Calculate the average speed of the cyclist between points E and F.
10. Calculate the average speed through the cyclist's entire journey.

Distance Versus Time Graph



The graph shows the journey of a woman taking the bus into town.

11. For the first part of her journey she walks to the bus stop. How long does this take?

12. How long did she wait for the bus to arrive?

13. How long does the journey take on the bus? (C to G)

14. How far from the woman's house is the bus stop?

15. How far from the woman's house is the town?

16. How fast does the woman walk to the bus stop?

17. What is the average speed of the entire bus journey?

18. If the woman were to walk to the town,
how long would it take her?

19. Suggest what is happening between points D and E.

Distance Versus Time Graph Answers

- How far has the cyclist traveled between points A and B?
30 m
- How far did the cyclist travel throughout his entire journey?
160 m
- Describe the cyclist's speed between points C and D.
Stationary / zero
- Describe the cyclist's speed between points D and E.
Moving quickly
- Describe the cyclist's speed between points B and C.
Moving slowly
- Between which two points is the cyclist traveling the fastest?
D and E
- Between which two points is the cyclist traveling the slowest?
B and C
- Calculate the average speed of the cyclist between points A and E.
4.44 m/s
- Calculate the average speed of the cyclist between points E and F.
11.43 m/s
- Calculate the average speed through the cyclist's entire journey.
6.40 m/s
- For the first part of her journey she walks to the bus stop. How long does this take?
180 s
- How long did she wait for the bus to arrive?
90 s
- How long does the journey take on the bus?
510 s
- How far from the woman's house is the bus stop?
400 m
- How far from the woman's house is the town?
2800 m
- How fast does the woman walk to the bus stop?
2.22 m/s
- What is the average speed of the entire bus journey?
4.71 m/s
- If the woman were to walk to the town, how long would it take her?
1261.26 s or 21.02 minutes
- Suggest what is happening between points D and E.
Stationary- another bus stop, traffic lights